

```
import pandas as pd
```

```
# Load the dataset to analyze it
```

```
file_path = '/content/shopping_trends.csv'
```

```
data = pd.read_csv(file_path)
```

```
# Display the first few rows of the dataset for inspection
```

```
data.head(), data.info()
```



```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 3900 entries, 0 to 3899
```

```
Data columns (total 19 columns):
```

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	object
3	Item Purchased	3900 non-null	object
4	Category	3900 non-null	object
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	object
7	Size	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3900 non-null	float64
11	Subscription Status	3900 non-null	object
12	Payment Method	3900 non-null	object
13	Shipping Type	3900 non-null	object
14	Discount Applied	3900 non-null	object
15	Promo Code Used	3900 non-null	object
16	Previous Purchases	3900 non-null	int64
17	Preferred Payment Method	3900 non-null	object
18	Frequency of Purchases	3900 non-null	object

```
dtypes: float64(1), int64(4), object(14)
```

```
memory usage: 579.0+ KB
```

(	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	\
0	1	55	Male	Blouse	Clothing	53	
1	2	19	Male	Sweater	Clothing	64	
2	3	50	Male	Jeans	Clothing	73	
3	4	21	Male	Sandals	Footwear	90	

	4	5	45	Male	Blouse	Clothing	49
	0	1	2	3	4		
	Location	Size	Color	Season	Review	Rating	Subscription Status \
0	Kentucky	L	Gray	Winter		3.1	Yes
1	Maine	L	Maroon	Winter		3.1	Yes
2	Massachusetts	S	Maroon	Spring		3.1	Yes
3	Rhode Island	M	Maroon	Spring		3.5	Yes
4	Oregon	M	Turquoise	Spring		2.7	Yes

	Payment Method	Shipping Type	Discount Applied	Promo Code Used \
0	Credit Card	Express	Yes	Yes
1	Bank Transfer	Express	Yes	Yes
2	Cash	Free Shipping	Yes	Yes
3	PayPal	Next Day Air	Yes	Yes
4	Cash	Free Shipping	Yes	Yes

	Previous Purchases	Preferred Payment Method	Frequency of Purchases
0	14	Venmo	Fortnightly
1	2	Cash	Fortnightly
2	23	Credit Card	Weekly
3	49	PayPal	Weekly
4	31	PayPal	Annually ,

None)

Question 1: Overall distribution of customer ages in the dataset

```
age_distribution = data['Age'].describe()
print(age_distribution)
```

```
count    3900.000000
mean      44.068462
std       15.207589
min       18.000000
25%       31.000000
50%       44.000000
75%       57.000000
max       70.000000
Name: Age, dtype: float64
```

Question 2: Average purchase amount by product category

```
avg_purchase_by_category = data.groupby('Category')['Purchase Amount (USD)'].mean()
print(avg_purchase_by_category)
```

```

Category
Accessories    59.838710
Clothing       60.025331
Footwear       60.255426
Outerwear      57.172840
Name: Purchase Amount (USD), dtype: float64

```

```

# Question 3: Gender with the highest number of purchases
gender_purchase_count = data['Gender'].value_counts()
print(gender_purchase_count)

```

```

Gender
Male      2652
Female    1248
Name: count, dtype: int64

```

```

# Question 4: Most commonly purchased items in each category
most_common_items_by_category = data.groupby('Category')['Item Purchased'].agg(
    lambda x: x.mode().iloc[0])
print(most_common_items_by_category)

```

```

Category
Accessories    Jewelry
Clothing       Blouse
Footwear       Sandals
Outerwear      Jacket
Name: Item Purchased, dtype: object

```

```

# Question 5: Seasons or months with significantly higher customer spending
season_spending = data.groupby('Season')['Purchase Amount (USD)'].sum()
print(season_spending)

```

```

Season
Fall      60018
Spring    58679
Summer    55777
Winter    58607
Name: Purchase Amount (USD), dtype: int64

```

```
# Question 6: Average rating for each product category
avg_rating_by_category = data.groupby('Category')['Review Rating'].mean()
print(avg_purchase_by_category)
```

```
Category
Accessories    59.838710
Clothing       60.025331
Footwear       60.255426
Outerwear      57.172840
Name: Purchase Amount (USD), dtype: float64
```

```
# Question 7: Differences in purchase behavior (subscribed vs. non-subscribed customers)
purchase_behavior_subscription = data.groupby('Subscription Status')['Purchase Amount (USD)'].mean()
print(purchase_behavior_subscription)
```

```
Subscription Status
No      59.865121
Yes     59.491928
Name: Purchase Amount (USD), dtype: float64
```

```
# Question 8: Most popular payment method
popular_payment_method = data['Payment Method'].value_counts()
print(popular_payment_method)
```

```
Payment Method
Credit Card    696
Venmo          653
Cash           648
PayPal         638
Debit Card     633
Bank Transfer  632
Name: count, dtype: int64
```

```
# Question 9: Spending differences between promo code users and non-users
spending_promo_code = data.groupby('Promo Code Used')['Purchase Amount (USD)'].mean()
print(spending_promo_code)
```

```
Promo Code Used
No      60.130454
Yes     59.279070
```

Name: Purchase Amount (USD), dtype: float64

Question 10: Frequency of purchases across age groups

```
age_bins = pd.cut(data['Age'], bins=[0, 18, 30, 45, 60, 100], labels=['0-18', '19-30', '31-45', '46-60', '60+'])
```

```
frequency_by_age_group = data.groupby(age_bins)['Frequency of Purchases'].value_counts(normalize=True)
```

```
print(frequency_by_age_group)
```

Age	Frequency of Purchases	
0-18	Every 3 Months	0.173913
	Quarterly	0.159420
	Annually	0.144928
	Bi-Weekly	0.144928
	Fortnightly	0.144928
	Weekly	0.130435
	Monthly	0.101449
19-30	Bi-Weekly	0.168182
	Quarterly	0.152273
	Every 3 Months	0.151136
	Fortnightly	0.144318
	Monthly	0.131818
	Annually	0.129545
	Weekly	0.122727
31-45	Annually	0.157365
	Every 3 Months	0.155535
	Monthly	0.153705
	Weekly	0.145471
	Fortnightly	0.140897
	Bi-Weekly	0.123513
	Quarterly	0.123513
46-60	Every 3 Months	0.149780
	Quarterly	0.145374
	Bi-Weekly	0.144493
	Monthly	0.144493
	Annually	0.139207
	Weekly	0.139207
	Fortnightly	0.137445
60+	Annually	0.163209
	Quarterly	0.163209
	Weekly	0.145228
	Every 3 Months	0.136929
	Monthly	0.135546
	Fortnightly	0.131397

Bi-Weekly 0.124481

Name: proportion, dtype: float64

```
<ipython-input-17-9ac832a5111a>:3: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version
frequency_by_age_group = data.groupby(age_bins)['Frequency of Purchases'].value_counts(normalize=True)
```

Question 11: Correlation between product size and purchase amount

```
size_purchase_correlation = data.groupby('Size')['Purchase Amount (USD)'].mean()
print(size_purchase_correlation)
```

```
Size
L      58.563153
M      59.924217
S      61.037707
XL     60.090909
Name: Purchase Amount (USD), dtype: float64
```

Question 12: Preferred shipping type by product category

```
preferred_shipping_by_category = data.groupby('Category')['Shipping Type'].agg(lambda x: x.mode().iloc[0])
print(preferred_shipping_by_category)
```

```
Category
Accessories      Store Pickup
Clothing          Standard
Footwear         Free Shipping
Outerwear        Free Shipping
Name: Shipping Type, dtype: object
```

Question 13: Effect of discounts on purchase decisions

```
discount_effect = data.groupby('Discount Applied')['Purchase Amount (USD)'].mean()
print(discount_effect)
```

```
Discount Applied
No      60.130454
Yes     59.279070
Name: Purchase Amount (USD), dtype: float64
```

Question 14: Most popular product colors

```
popular_colors = data['Color'].value_counts()
```

```
print(popular_colors)
```

```
→ Color
Olive      177
Yellow     174
Silver     173
Teal       172
Green      169
Black      167
Cyan       166
Violet     166
Gray       159
Maroon     158
Orange     154
Charcoal   153
Pink       153
Magenta    152
Blue       152
Purple     151
Peach      149
Red        148
Beige      147
Indigo     147
Lavender   147
Turquoise  145
White      142
Brown      141
Gold       138
Name: count, dtype: int64
```

```
# Question 15: Average number of previous purchases by customers
```

```
avg_previous_purchases = data['Previous Purchases'].mean()
```

```
print(avg_previous_purchases)
```

```
→ 25.35153846153846
```

```
# Question 16: Purchase amount differences based on review ratings
```

```
purchase_by_ratings = data.groupby('Review Rating')['Purchase Amount (USD)'].mean()
```

```
print(purchase_by_ratings)
```

```

Review Rating
2.5    62.287879
2.6    59.566038
2.7    59.363636
2.8    57.066176
2.9    56.470588
3.0    60.728395
3.1    58.770701
3.2    61.315789
3.3    59.861842
3.4    59.005495
3.5    58.833333
3.6    57.322148
3.7    58.974359
3.8    60.873239
3.9    58.926380
4.0    59.237569
4.1    61.959459
4.2    60.853801
4.3    59.673469
4.4    60.525316
4.5    59.489209
4.6    57.683908
4.7    59.283784
4.8    61.881944
4.9    63.885542
5.0    64.352941
Name: Purchase Amount (USD), dtype: float64

```

```

# Question 17: Differences in purchase behavior between locations
purchase_by_location = data.groupby('Location')['Purchase Amount (USD)'].mean()
print(purchase_by_location)

```

```

Location
Alabama    59.112360
Alaska     67.597222
Arizona    66.553846
Arkansas   61.113924
California 59.000000
Colorado   56.293333
Connecticut 54.179487
Delaware   55.325581

```


Florida	55.852941
Georgia	58.797468
Hawaii	57.723077
Idaho	60.075269
Illinois	61.054348
Indiana	58.924051
Iowa	60.884058
Kansas	54.555556
Kentucky	55.721519
Louisiana	57.714286
Maine	56.987013
Maryland	55.755814
Massachusetts	60.888889
Michigan	62.095890
Minnesota	56.556818
Mississippi	61.037500
Missouri	57.913580
Montana	60.250000
Nebraska	59.448276
Nevada	63.379310
New Hampshire	59.422535
New Jersey	56.746269
New Mexico	61.901235
New York	60.425287
North Carolina	60.794872
North Dakota	62.891566
Ohio	60.376623
Oklahoma	58.346667
Oregon	57.337838
Pennsylvania	66.567568
Rhode Island	61.444444
South Carolina	58.407895
South Dakota	60.514286
Tennessee	61.974026
Texas	61.194805
Utah	62.577465
Vermont	57.176471
Virginia	62.883117
Washington	63.328767
West Virginia	63.876543
Wisconsin	55.946667
Wyoming	60.690141

Name: Purchase Amount (USD), dtype: float64

```
# Question 18: Relationship between age and product category purchased
age_category_relationship = data.groupby(age_bins)['Category'].value_counts(normalize=True)
print(age_category_relationship)
```

```
Age      Category      proportion
0-18     Clothing      0.478261
         Accessories    0.260870
         Outerwear      0.144928
         Footwear       0.115942
19-30     Clothing      0.457955
         Accessories    0.319318
         Footwear       0.146591
         Outerwear      0.076136
31-45     Clothing      0.447392
         Accessories    0.333943
         Footwear       0.139067
         Outerwear      0.079597
46-60     Clothing      0.433480
         Accessories    0.293392
         Footwear       0.186784
         Outerwear      0.086344
60+       Clothing      0.442600
         Accessories    0.336100
         Footwear       0.135546
         Outerwear      0.085754
```

Name: proportion, dtype: float64

```
<ipython-input-25-7d6a52ebb89a>:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future ve
age_category_relationship = data.groupby(age_bins)['Category'].value_counts(normalize=True)
```

```
# Question 19: Average purchase amount between male and female customers
purchase_by_gender = data.groupby('Gender')['Purchase Amount (USD)'].mean()
print(purchase_by_gender)
```

```
Gender
Female    60.249199
Male      59.536199
Name: Purchase Amount (USD), dtype: float64
```

```
final_results = {
    "Age Distribution": age_distribution
```

```

Age Distribution : age_distribution,
"Average Purchase by Category": avg_purchase_by_category,
"Gender with Most Purchases": gender_purchase_count,
"Most Common Items by Category": most_common_items_by_category,
"Seasonal Spending": season_spending,
"Average Rating by Category": avg_rating_by_category,
"Purchase Behavior (Subscribed vs Non-Subscribed)": purchase_behavior_subscription,
"Most Popular Payment Method": popular_payment_method,
"Promo Code Spending Differences": spending_promo_code,
"Frequency by Age Group": frequency_by_age_group,
"Size and Purchase Amount Correlation": size_purchase_correlation,
"Preferred Shipping by Category": preferred_shipping_by_category,
"Discount Effect": discount_effect,
"Most Popular Colors": popular_colors,
"Average Previous Purchases": avg_previous_purchases,
"Purchase by Ratings": purchase_by_ratings,
"Purchase by Location": purchase_by_location,
"Age and Product Category Relationship": age_category_relationship,
"Purchase Amount by Gender": purchase_by_gender
}

```

final_results

```

↔ {'Age Distribution': count      3900.000000
   mean      44.068462
   std       15.207589
   min       18.000000
   25%       31.000000
   50%       44.000000
   75%       57.000000
   max       70.000000
   Name: Age, dtype: float64,
   'Average Purchase by Category': Category
   Accessories      59.838710
   Clothing          60.025331
   Footwear         60.255426
   Outerwear        57.172840
   Name: Purchase Amount (USD), dtype: float64,
   'Gender with Most Purchases': Gender
   Male            2652
   Female         1248
   Name: count, dtype: int64,

```

```

'Most Common Items by Category': Category
Accessories      Jewelry
Clothing         Blouse
Footwear         Sandals
Outerwear        Jacket
Name: Item Purchased, dtype: object,
'Seasonal Spending': Season
Fall            60018
Spring         58679
Summer         55777
Winter          58607
Name: Purchase Amount (USD), dtype: int64,
'Average Rating by Category': Category
Accessories      3.768629
Clothing         3.723143
Footwear         3.790651
Outerwear        3.746914
Name: Review Rating, dtype: float64,
'Purchase Behavior (Subscribed vs Non-Subscribed)': Subscription Status
No              59.865121
Yes             59.491928
Name: Purchase Amount (USD), dtype: float64,
'Most Popular Payment Method': Payment Method
Credit Card     696
Venmo            653
Cash            648
PayPal          638
Debit Card      633
Bank Transfer   632
Name: count, dtype: int64,
'Promo Code Spending Differences': Promo Code Used
No              60.130454
Yes             59.279070
Name: Purchase Amount (USD), dtype: float64,
'Frequency by Age Group': Age      Frequency of Purchases
0-18  Every 3 Months      0.173913
      Quarterly          0.159420
      Annually           0.144928
      Bi-Weeklv         0.144928

```

```
import matplotlib.pyplot as plt
```

```
# Define questions for graphs and tabular outputs
```

```
questions_to_plot = {
    "Age Distribution": age_distribution,
    "Average Purchase by Category": avg_purchase_by_category,
    "Gender with Most Purchases": gender_purchase_count,
    "Seasonal Spending": season_spending,
    "Subscribed vs Non-Subscribed Behavior": purchase_behavior_subscription,
    "Most Popular Payment Method": popular_payment_method[:5], # Top 5 payment methods
    "Promo Code Spending Differences": spending_promo_code,
    "Popular Colors": popular_colors[:10], # Top 10 colors
}

# Plotting the data
fig, axs = plt.subplots(4, 2, figsize=(16, 18))
axs = axs.flatten()

for idx, (title, data) in enumerate(questions_to_plot.items()):
    if isinstance(data, pd.Series):
        data.plot(kind="bar", ax=axs[idx], color='skyblue', edgecolor='black')
    else: # For non-series data, we skip plotting
        continue
    axs[idx].set_title(title, fontsize=14)
    axs[idx].set_ylabel("Value")
    axs[idx].set_xlabel("Categories" if idx != 0 else "Age")

# Remove unused subplots
for i in range(len(questions_to_plot), len(axs)):
    fig.delaxes(axs[i])

plt.tight_layout()
plt.show()

# Tabular outputs
tables = {
    "Average Purchase by Category": avg_purchase_by_category,
    "Seasonal Spending": season_spending,
    "Subscribed vs Non-Subscribed Behavior": purchase_behavior_subscription,
    "Promo Code Spending Differences": spending_promo_code,
    "Popular Colors (Top 10)": popular_colors[:10],
}
```

tables

